Experimental Economics
https://doi.org/10.1007/s10683-020-09648-w

ORIGINAL PAPER

# The relative income effect: an experiment

**John Ifcher[1] · Homa Zarghamee[2] · Dan Houser[3] · Lina Diaz[3]**

Received: 20 February 2019 / Revised: 6 February 2020 / Accepted: 14 February 2020
© Economic Science Association 2020

## Abstract

John Stuart Mill claimed that "men do not desire merely to be rich, but richer than other men." Are people made happy by being richer than others? Or are people made happy by favorable comparisons to others more generally, and being richer is merely a proxy for this ineffable relativity? We conduct an online experiment absent choice in which we measure subjective well-being (SWB) before and after an exogenous shock that reveals to subjects how many experimental points they and another subject receive, and whether or not points are worth money. We find that subjects are made significantly happier when they receive monetized rather than non-monetized points, suggesting money is valued more than the points it represents. In contrast, subjects are made equally unhappy when they receive fewer monetized points as when they receive fewer non-monetized points than others, suggesting relative money is not valued more than the relative points it represents. We find no evidence that subjects are made happier by being "richer" than others (i.e., by receiving more points—either monetized or non-monetized—than others). Subgroup analyses reveal women are made unhappier (than men) by being "richer" and "poorer" than others, and conservatives are made unhappier (than progressives) by being "poorer" than others. Our experimental-SWB approach is easy to administer and may complement choice-based tasks in future experiments to better estimate preference parameters.

**Keywords** Subjective well-being · Relative income · Others' income · Income comparisons · Happiness · Experiments

**JEL Classification** C91 · D31 · D63 · I31

**Electronic supplementary material** The online version of this article (https://doi.org/10.1007/s10683-020-09648-w) contains supplementary material, which is available to authorized users.

✉ Homa Zarghamee
hzargham@barnard.edu

[1] Santa Clara University, Santa Clara, CA, USA

[2] Barnard College, New York, NY, USA

[3] George Mason University, Fairfax, VA, USA

Published online: 11 March 2020

## 1 Introduction

John Stuart Mill is credited with the claim that "men do not desire merely to be rich, but richer than other men."[1] This claim raises two fundamental questions. Does being richer than others make people happy? And if so, why? Empirical evidence regarding the first question is mixed. While subjective well-being (SWB) scholars have assembled substantial empirical evidence from large observational datasets of a negative relationship between others' income and one's own SWB (Blanchflower and Oswald 2004; Clark et al. 2008; Helliwell et al. 2012; Luttmer 2005), identification is often confounded. For example, some studies have found that cost-of-living explains the negative relationship (e.g., Ifcher et al. 2018), while others have estimated a positive relationship (Brodeur and Fleche 2019; Clark et al. 2009a; Deaton and Stone 2013; D'Ambrosio and Frick 2012; FitzRoy et al. 2014; Ifcher et al. 2018; Kingdon and Knight 2007; Senik 2004, 2008). A simple answer to the second question is that people care about relative consumption—that is, they are made happy by consuming more than others. Another plausible explanation that has not been explored in the SWB literature, though, is that people are made happy by comparing favorably to others in a more general sense, and being richer than others proxies for this ineffable relativity. In this case, being richer than others may only appear to increase happiness in the absence of other measures of relativity.

In this paper, we attempt to address these two questions using an experiment. Specifically, we measure subjects' SWB before and after an exogenous relative-rewards shock in which subjects learn the number of experimental points they and another subject receive (2 or 10), and whether each experimental point is monetized (worth \$1) or non-monetized (worth \$0). This design enables us to compare the SWB-change of two subjects receiving the same experimental points who differ only in the points they learn another subject receives. Comparison of the impact of monetized and non-monetized points allows us to identify the impact of others' money on one's SWB and to determine whether non-monetized points generate the same observed relativity. To illustrate the nuance of this approach, consider a less nuanced version wherein a subject learns that she receives \$2 and another subject receives \$10. SWB may change because the subject learns that she is receiving \$8 less than the other subject, but it may also be affected by learning that: (1) she is receiving \$2; (2) the other subject is receiving \$10; (3) there is a metric on which she is in worse relative standing than the other subject by 8 units; or (4) she is receiving \$4 less than she expected when agreeing to participate in the experiment. Our design attempts to rule out explanations (1)–(4).

First, we find that it is not SWB-improving to learn that the other subject receives a smaller—in comparison to an equal—number of monetized points, suggesting that subjects are not made happier by being "richer" than other subjects *ceteris paribus*. Second, we find that while it is SWB-diminishing to learn that the other subject receives a larger—in comparison to an equal—number of monetized points, it

[1] Pigou (1920) attributes this quotation to Mill (Luttmer 2005), but its authorship is contested by Mill scholars [see Rees (1956)]

is statistically indistinguishably SWB-diminishing when points are non-monetized. This suggests that others' money impacts own SWB only insofar as it proxies for others' points; that is, subjects are only made less happy about being "poorer" because it means they receive fewer points than others, not because they receive less money than others. In contrast, subjects do seem to be made happier by receiving money themselves, and not just for the points that the money represents: while it is SWB-improving to receive non-monetized points, it is significantly more SWB-improving to receive monetized points.

Our research is also motivated by the increasing use of SWB measures in public policy. In an influential critique of GDP as the singular measure of economic health and well-being, the Commission on the Measurement of Economic Performance and Social Progress recommended that policy makers consider SWB data (Stiglitz et al. 2010). Researchers have since been trying to understand the nature and limits of SWB indices as measures of well-being suitable for policy (e.g., Benjamin et al. 2019). For example, Benjamin et al. (2012) conduct an experiment in which subjects read scenarios, predict their SWB in the scenarios, and make hypothetical choices between the scenarios; Benjamin et al. (2014a) compares medical students' real rankings of residency programs with their predicted SWB in those programs; and Benjamin et al. (2014b) estimates marginal utilities of various SWB measures and aspects of well-being. By testing whether and which SWB measures are responsive to experimental rewards, our experiment furthers inquiry into whether the experimental setting can offer for SWB policy researchers the combination of measurable outcomes and control that it offers choice-based researchers; as in all experiments, such lab-appropriate measures—along with any relativity that we identify in the experimental setting—should not be assumed to abide by the same dynamics observed in survey-based data.

## 2 Literature review

The use of SWB data by researchers and policy-makers has proliferated in recent years: several national governments (e.g., Britain, France, and Bhutan) now collect SWB data from their citizens, and the World Bank now compiles SWB research and statistics in an annual World Happiness Report. SWB measures are numerical indices of respondents' sense of well-being and can range from items like, agreement with the statement "Did you feel worried for a lot of the day yesterday?" to agreement with "I lead a purposeful and meaningful life." Indeed, researchers have classified three broad categories of SWB measures: experiential-SWB measures assess feelings, experiences, or emotions over short time-frames, like the present moment or in the last week; evaluative-SWB measures assess overall or domain-specific satisfaction or happiness with life; and eudemonic-SWB measures assess the fulfillment of underlying psychological needs, like respect, autonomy, and self-acceptance. The three measures are sometimes similarly responsive to potential correlates of SWB and sometimes not. For example, while a positive cross-sectional relationship between income and all three dimensions of SWB is well-established in the literature (e.g., Stevenson and Wolfers 2008; Kahneman and Deaton 2010; Clark

and Senik 2011), Kahneman and Deaton (2010) find that the relationship has a satiation point of $75,000 when using experiential but not evaluative measures; Stone and Mackie (2013) find that the relationship between SWB and having children is negative using experiential measures but positive using evaluative measures. Dolan et al. (2011) discusses the policy-importance of separately measuring experiential, evaluative, and eudemonic SWB.

Often using evaluative measures, but also with experiential measures, SWB scholars have found evidence of a negative relationship between others' income and one's own SWB that is attributed to a "relative income effect" (RIE): income comparisons cause SWB to decrease with others' income, *ceteris paribus*.[2] The RIE has important implications, for example, regarding the potential benefits of economic growth. Some SWB researchers believe that the RIE helps explain the Easterlin Paradox: the empirical observation that over time in many countries, average national SWB does not increase with real per capita GDP (Easterlin 1974, 2013; Easterlin et al. 2010). Luttmer (2005) reports a negative relationship between regional median income and SWB that is as big as the positive relationship between own income and SWB, implying that shared economic growth would not be associated with improved national SWB.

Identification of the RIE, though, is often confounded. First, observational studies do not generally account for selection (e.g., into neighborhoods or occupations). Indeed, the studies with the most credible exogenous relative-income shocks—the Moving to Opportunities for Fair Housing demonstration and the Dutch Postcode Lottery—do not find a negative effect of neighbors' income on one's SWB (Ludwig et al. 2013; Kuhn et al. 2011). Second, *ceteris paribus* is easily violated in observational studies. Ifcher et al. (2018) find that adding controls for cost-of-living renders the significant negative relationship between others' income and own SWB insignificant, suggesting that the original significant relationship could be explained by cost-of-living. Further, the sign of the others-income-own-SWB relationship is positive in some contexts. A positive relationship has been identified in immediate neighborhoods and has been attributed to local public goods (Ifcher et al. 2018; Brodeur and Fleche 2019; Deaton and Stone 2013;[3] Ludwig et al. 2013; Clark et al. 2009b) or to altruism (Kingdon and Knight 2007). A positive relationship has also been identified in periods of rapid economic growth and has been attributed to "the tunnel effect," whereby others' income serves as a signal of one's future income (Hirschman and Rothschild 1973; Senik 2004, 2008).

These violations of *ceteris paribus* illustrate that others' income can impact SWB through channels other than the RIE. They do not, however, prove or disprove the existence of the RIE itself, as the sign of the relationship between others' income

[2] What we here call the RIE should not be mistaken for the Relative Income Hypothesis (RIH). While the RIE has SWB as its outcome variable, RIH, originally put forth by Duesenberry (1949), is the theory that consumption and savings *behavior* depend on one's income level relative to others'. It is technically ambiguous whether our motivating quote from Mill relates more to RIE or RIH. We speak further to the relationship between SWB and behavior in the discussion.

[3] Deaton and Stone (2013) estimate a negative relationship using experiential measures and a positive relationship using evaluative measures.

and own SWB represents a net effect of an indeterminate set of channels. For example, Clark et al. (2009b) find that, controlling for own income, neighborhood median income and own SWB are positively correlated, which the authors attribute to local public goods. At the same time, controlling for own income and neighborhood median income, they find that one's income-rank within one's neighborhood is positively correlated with own SWB, which they attribute to the RIE.

These confounds render compelling the control afforded by an experimental analysis. To our knowledge, the only such experiment with real-money rewards is McBride (2010), who attempts to identify the RIE in a single-player, multiple-round, penny-matching game played against a computer. In each round, subjects are first informed of the computer's randomized probability of choosing heads or tails (there are five possible types), and then they choose heads or tails. Next, subjects learn their own payment in that round and, depending on treatment, either: (1) no further information, (2) the average payment of all other subjects, or (3) the average payment of subjects by computer-probability-type. Subjects then report their satisfaction with the results of the round. Satisfaction with a round's results significantly decreases with the average payment of other subjects in (2) and only with the average payment of subjects with the same probability-type in (3). While these results appear consistent with the RIE, identification is confounded because subjects have agency: their satisfaction (dissatisfaction) may not be due to income per se but to feelings of relative success or self-congratulations (failure or self-criticism).[4]

## 3 Experimental design

### 3.1 Discovery and pilot studies

Like the main online experiment, the goal of the discovery study was to identify the SWB-impact of an exogenous relative-rewards shock. A more general ambition was to determine whether the control afforded by laboratory experimentation could be leveraged to study SWB (as measured in large observational datasets and increasingly in national accounts) and its determinants, especially income. The discovery study was necessary to determine SWB measures' sensitivity to money-rewards of the size paid in experiments. This required testing the many different measures of SWB available in the literature, and varying payment-sizes. As such, the discovery study, while similar in methodology to the online experiment, includes much longer

[4] It warrants mention that other experiments have found evidence supportive of the RIE using choices over hypothetical scenarios (Johansson-Stenman et al. 2002; Solnick and Hemenway 1998; Zeckhauser 1991). Also supportive of the RIE, Smith et al. (1989) document higher SWB with a given hypothetical wage when it is from a distribution with a lower mean. Further, there are three economic experiments in which relative real-money payments vary and SWB is measured. The correlation between others' payments and subjects' SWB is consistent with the RIE in Bosman and van Winden (2002) and Konow and Earley (2008) and inconsistent with the RIE in Charness and Grosskopf (2001). That said, these experiments are not designed to test the RIE so attempts to attribute the correlations to the RIE are beset by endogeneity and other identification issues.

SWB surveys and more rewards-shocks. Details and discussion of the discovery study are presented in Online Appendix A.

While the discovery study enabled us to refine the SWB survey and revelation mechanism, the pilot study enabled us to test the revised methodology and help estimate the number of subjects that were needed for the online experiment. Online Appendix A also includes details and discussion of the pilot study.[5]

### 3.2 Online experiment

To study the RIE, we conducted an online experiment in December 2017. Prospective subjects were recruited using the online crowdsourcing marketplace Amazon Mechanical Turk (mTurk). The set of potential subjects consisted of mTurk "workers"—people seeking to make money by performing tasks on the platform—whose accounts were registered in the United States and had at least 95% of their past tasks approved by employers. The description of the task on mTurk informed potential workers (subjects) that participation would take less than 20 min, that they would be paid for their participation (minimum, average, and maximum payments of $3.00, $7.00, and $11.00), and that the payment-amount would be determined randomly. Each recruited subject received a link to Qualtrics, where she received an exogenous relative-rewards shock (revelation) and completed pre- and post-revelation SWB surveys, allowing for measurement of the SWB-effect of the revelation. Subjects entered a survey-completion code generated by Qualtrics into mTurk to receive their payments. 996 subjects completed the experiment in 10 min on average and were paid a minimum, average, and maximum of $3.00, $7.27, and $11.00.[6] Screenshots are included in Online Appendix B.

#### 3.2.1 Pre- and post-revelation SWB surveys

The pre- and post-revelation SWB surveys each included the Mood Short Form (MSF), which enables a quick measurement of subjects' experiential SWB (Peterson and Sauber 1983). The MSF includes the following four items, with a five-point Likert response-scale for each item ("Strongly disagree" = 1, "Disagree" = 2, "Neither agree nor disagree" = 3, "Agree" = 4, and "Strongly agree" = 5):

"Currently I am in a good mood."
"As I answer these questions, I feel very cheerful."
"For some reason I am not very comfortable."
"At this moment I feel 'edgy' or irritable."

[5] Columns (1) and (2) of Appendix Table 7 present the results of pooling the pilot study and online experiments, using the methods presented in Sect. 4.2 below; the results are materially unchanged.

[6] Of 1180 individuals who clicked the Qualtrics link, 128 did not complete the survey, and another 56 completed the survey but either entered an incorrect survey-completion code or none at all. These individuals could not be paid and are not included as subjects in our analyses.

MSF scores are calculated by summing the four responses with the response-scale reverse-coded for the last two items. Possible scores range from 4 (worst possible mood) to 20 (best possible mood).

In an attempt to reduce the propensity of subjects to anchor their post- to pre-revelation MSF responses, items from the Basic Psychological Need Satisfaction Scale (BPNSS) were included in both surveys. The BPNSS is a 21-item instrument that measures needs for competence, autonomy, and relatedness (Deci and Ryan 2000; Gagné 2003). Responders evaluate statements (e.g., "Often, I do not feel very confident," and "People in my life care about me") on a seven-point scale in which "Not at all true" = 1, "Somewhat true" = 4, and "Very true" = 7. The pre- and post-revelation SWB surveys included 11 and 10 BPNSS items, respectively.

To further reduce the propensity to anchor, the order of all items in the pre-revelation SWB survey was randomized by subject. The post-revelation items were randomized in two strata: the first contained the four MSF items randomized by subject, and the second contained the 10 BPNSS items also randomized by subject. The stratified randomization ensured that the post-revelation MSF items were presented immediately after the revelation, thus reducing concerns that the impact of the revelation may have worn off.

While Online Appendix A details and discusses the discovery study, it is important to note why we use the MSF in the pilot and online experiments. For the discovery study, we compiled a comprehensive list of SWB items and scales that had been previously used in the SWB literature, including eudemonic, evaluative, and experiential measures; these came from oft-cited SWB datasets (e.g., the World Values Survey), and contained popular SWB measures (e.g., Cantril's Ladder, a single-item measure) and scales (e.g., Positive Affect Negative Affect Scale, a multi-item measure). In the discovery sample, subjects completed a total of 92 SWB items (8 single-item and 7 multi-item measures), and, of these, the MSF was found to be the most responsive to revelation and the BPNSS among the least responsive.

### 3.2.2 Revelation mechanism

The revelation mechanism was designed to provide an unconfounded relative-rewards shock. Prior to the pre-revelation SWB survey, in an attempt to standardize expectations, the instructions fully informed subjects about the revelation mechanism. Each subject was informed that:

- She would be paid a $1 reward for completing the study.[7]
- She would be randomly assigned to a two-person group.
- The other subject in the group (hereafter Participant X) could be any other subject in the study.

[7] In mTurk, a flat payment that all subjects receive is called a "reward;" and a payment that can vary by subject is called a "bonus." It should be noted that our payments are sizeable for mTurk, where workers' median hourly rewards have been estimated to be $2, and only 4% of workers earn more than $7.25 per hour (Hara et al. 2018).

- She would never learn Participant X's identity and vice versa.
- She and Participant X would be allotted 2 or 10 experimental points each, creating four possible allocations of points:
  - She receives 2 points, and Participant X receives 2 points.
  - She receives 2 points, and Participant X receives 10 points.
  - She receives 10 points, and Participant X receives 2 points.
  - She receives 10 points, and Participant X receives 10 points.

Below, we refer to these allocations as low–low (LL), low–high (LH), high–low (HL), and high–high (HH), respectively.

Subjects were then randomly informed of being in one of two treatments. Each subject in the "points-money (pts$) treatment" was informed that each experimental point was worth $1. Each subject in the "points (pts) treatment" was informed that she and Participant X would receive a $6 payment regardless of the number of experimental points. Subjects were presented with neither the allotment nor the treatment terminology. Also, subjects in the pts$-treatment did not know about the pts-treatment, and vice versa.

After completing the pre-revelation SWB survey, subjects were reminded, as appropriate, that either "the number of points you and Participant X receive will not affect your bonus payments," or "you and Participant X will receive a bonus payment of a dollar for each point you receive." Subjects were also reminded that they were randomly assigned to one of the four possible allocations of points. To reveal the allocation of points, subjects had to press an "OK" button. They were then informed of the number of points they and Participant X received.

#### 3.2.3 Screening questions, questionnaire, and end of experiment

Though mTurk subjects have been shown to be more attentive than convenience samples from universities (Hauser and Schwarz 2016), because our survey items were not individually incentivized, the experiment included four screening questions to allay any concerns about workers being inattentive or having programmed responses:

- After reading the instructions and before answering the pre-revelation SWB survey, subjects were asked what year it was; there were five possible responses: "1990," "2017," "2000," "2018," and "2019."
- After completing the pre-revelation SWB survey, subjects were presented with the following statement: "Currently, the year is 2025." The response scale was the same five-point Likert scale used with the MSF items.[8]

[8] To minimize the impact of these screening questions on subjects' MSF responses, we chose to use questions about what year it was, as the questions' content is affectively neutral. We asked the two questions in two different ways to encourage attention, and the second question was designed as a statement with a Likert scale to mimic MSF items.

**Table 1** 4X2 treatment-design, with own, Participant X's, and relative rewards by cell

| Cell | pts-$-Treatment | | | pts-Treatment | | |
|---|---|---|---|---|---|---|
| | Self | Participant X | Relative rewards | Self | Participant X | Relative rewards |
| LL | $2 | $2 | 0 | $6 | $6 | 0 |
| | 2 pts | 2 pts | 0 | 2 pts | 2 pts | 0 |
| LH | $2 | $10 | −$8 | $6 | $6 | 0 |
| | 2 pts | 10 pts | −8 pts | 2 pts | 10 pts | −8 pts |
| HL | $10 | $2 | +$8 | $6 | $6 | 0 |
| | 10 pts | 2 pts | +8 pts | 10 pts | 2 pts | +8 pts |
| HH | $10 | $10 | 0 | $6 | $6 | 0 |
| | 10 pts | 10 pts | 0 | 10 pts | 10 pts | 0 |

As the instructions disclosed all possible revelations, the subject's expected pts- and $-rewards are 6 pts and $6 for both herself and Participant X for all cells. Relative rewards are own minus Participant X's rewards.

- In the post-revelation SWB survey (after completing the MSF items and before completing the BPNSS items), subjects completed the following two items:
  - "Please indicate the value of the bonus you will receive (in addition to the $1 reward you will receive for completing this study)." Possible responses were "$2," "$6," and "$10."
  - "Please indicate the value of the bonus Participant X will receive (in addition to the $1 reward Participant X will receive for completing this study)." Possible responses were "$2," "$6," and "$10."

After the post-revelation SWB survey, subjects completed a 15-item questionnaire (including demographic characteristics and political views), entered their survey-completion codes, and were paid.

### 3.3 Own- and relative-rewards shocks

Our experiment yields 8 distinct cells from a 4 (allocations of 2 or 10 pts to the subject and 2 or 10 pts to Participant X) X 2 (pts$- vs. pts-treatment) design: $LL_{pts\$}$, $LH_{pts\$}$, $HL_{pts\$}$, $HH_{pts\$}$, $LL_{pts}$, $LH_{pts}$, $HL_{pts}$, and $HH_{pts}$ (see Table 1). This design enables identification of both relative-rewards and own-rewards shocks by comparing cells that hold all other factors constant.

For example, let's compare the HL and HH cells by treatment. Each subject in the $HL_{pts\$}$ cell was told that she would receive 10 pts worth $1 each and that Participant X would receive 2 pts worth $1 each, whereas each subject in the $HH_{pts\$}$ cell was told that she and Participant X would each receive 10 pts worth $1 each. In these two cells, each subject's realized pts- and $-rewards are constant (10 pts and $10). Further, as the instructions disclosed all possible revelations, each subject's expected pts- and $-rewards should be 6 pts and $6 for both herself and Participant X. Thus, the $HL_{pts\$}$ and $HH_{pts\$}$ cells differ only in Participant X's realized pts- and $-rewards (2 pts and $2 vs. 10 pts and $10). By comparing the MSF-change

(post- minus pre-revelation) of $HL_{pts\$}$ and $HH_{pts\$}$ subjects, we can identify *ceteris paribus* the differential impact on a subject's mood of Participant X receiving 2 pts and \$2 rather than 10 pts and \$10: $\Delta MSF_{HL,pts\$} - \Delta MSF_{HH,pts\$}$. We refer to this as the impact of an *advantageous* relative-pts-\$ shock. In relative-rewards-shock comparisons, "advantageous" indicates that the subject receives greater rewards than Participant X, rather than the same rewards as Participant X.

Let's now consider the analogous cells of the pts-treatment. Because pts are not monetized, and the realized \$-rewards are constant in the $HL_{pts}$ and $HH_{pts}$ cells, $\Delta MSF_{HL,pts} - \Delta MSF_{HH,pts}$ identifies *ceteris paribus* the differential impact on a subject's mood of Participant X receiving 2 pts rather than 10 pts. We refer to this as the impact of an advantageous relative-pts shock. Therefore, we can capture the impact of Participant X receiving \$2 rather than \$10 by subtracting the impact of an advantageous relative-pts shock from the impact of an advantageous relative-pts-\$ shock: $(\Delta MSF_{HL,\$} - \Delta MSF_{HH,\$}) = (\Delta MSF_{HL,pts\$} - \Delta MSF_{HH,pts\$}) - (\Delta MSF_{HL,pts} - \Delta MSF_{HH,pts})$. We refer to this difference as the impact of an advantageous relative-\$ shock.

In relative-rewards-shock comparisons, "disadvantageous" indicates that the subject receives lesser rewards than Participant X, rather than the same rewards as Participant X. By comparing LH and LL cells, we can identify the impact of disadvantageous relative-pts-\$ ($\Delta MSF_{LH,pts\$} - \Delta MSF_{LL,pts\$}$), relative-pts ($\Delta MSF_{LH,pts} - \Delta MSF_{LL,pts}$), and relative-\$ ($(\Delta MSF_{LH,\$} - \Delta MSF_{LL,\$}) = (\Delta MSF_{LH,pts\$} - \Delta MSF_{LL,pts\$})$—$(\Delta MSF_{LH,pts} - \Delta MSF_{LL,pts})$) shocks.

Similarly, we can identify the impact of an *advantageous* own-pts-\$ shock by comparing $HH_{pts\$}$ to $LL_{pts\$}$ cells: $\Delta MSF_{HH,pts\$} - \Delta MSF_{LL,pts\$}$. Subjects in the $HH_{pts\$}$ ($LL_{pts\$}$) cells are told that they and Participant X will each receive 10 pts (2 pts) worth \$1 each. In both cells, subjects and Participant X receive the same rewards, thus the RIE should not explain the impact of the advantageous own-pts-\$ shock. Therefore, in own-rewards-shock comparisons, "advantageous" indicates that the subject receives greater rewards than expected, rather than lesser rewards than expected (recall that expected rewards are 6 pts and \$6 for all subjects).[9]

Comparing the analogous cells of the pts-treatment, we can identify the impact of an advantageous own-pts shock by comparing $HH_{pts}$ and $LL_{pts}$ cells: $\Delta MSF_{HH,pts} - \Delta MSF_{LL,pts}$. Because pts are not monetized and the realized \$-rewards are constant, subjects in these cells differ only in the pts-rewards that they and Participant X receive (10 pts each versus 2 pts each). Therefore, we can capture the impact of the subject receiving \$10 rather than \$2 by subtracting the impact of an advantageous own-pts shock from the impact of an advantageous own-pts-\$ shock: $(\Delta MSF_{HH,\$} - \Delta MSF_{LL,\$}) = (\Delta MSF_{HH,pts\$} - \Delta MSF_{LL,pts\$}) - (\Delta MSF_{HH,pts} - \Delta MSF_{LL,pts})$. We refer to this difference as the impact of an advantageous own-\$ shock.

[9] This interpretation of the impact of an advantageous own-pts-\$ shock is potentially confounded if Participant X's rewards impact subjects' SWB through channels other than the RIE (e.g., if subjects are social surplus maximizers). This is because, while subjects' rewards relative to Participant X are the same in the HHpts\$ and LLpts\$ cells, Participant X's absolute rewards vary.

**Table 2** Relative- and own-rewards shocks and formulae used for estimation

| A. Shocks | B. Difference-of-means estimators | C. Regression estimators |
|---|---|---|
| Advantageous relative-pts-$ | $\Delta MSF_{HL,pts\$} - \Delta MSF_{HH,pts\$}$ | $\boldsymbol{-(\beta_4+\beta_6+\beta_7)} = (\beta_0+\beta_2+\beta_3) - (\beta_0+\beta_2+\beta_3+\beta_4+\beta_6+\beta_7)$ |
| Advantageous relative-pts | $\Delta MSF_{HL,pts} - \Delta MSF_{HH,pts}$ | $\boldsymbol{-(\beta_4+\beta_7)} = (\beta_0+\beta_3) - (\beta_0+\beta_3+\beta_4+\beta_7)$ |
| Advantageous relative-$ | $(\Delta MSF_{HL,pts\$} - \Delta MSF_{HH,pts\$}) - (\Delta MSF_{HL,pts} - \Delta MSF_{HH,pts})$ | $\boldsymbol{-\beta_6} = -(\beta_4+\beta_6+\beta_7) + (\beta_4+\beta_7)$ |
| Disadvantageous relative-pts-$ | $\Delta MSF_{LH,pts\$} - \Delta MSF_{LL,pts\$}$ | $\boldsymbol{\beta_4+\beta_5} = (\beta_0+\beta_1+\beta_4+\beta_5) - (\beta_0+\beta_1)$ |
| Disadvantageous relative-pts | $\Delta MSF_{LH,pts} - \Delta MSF_{LL,pts}$ | $\boldsymbol{\beta_4} = (\beta_0+\beta_4) - \beta_0$ |
| Disadvantageous relative-$ | $(\Delta MSF_{LH,pts\$} - \Delta MSF_{LL,pts\$}) - (\Delta MSF_{LH,pts} - \Delta MSF_{LL,pts})$ | $\boldsymbol{\beta_5} = (\beta_4+\beta_5) - \beta_4$ |
| Advantageous own-pts-$ | $\Delta MSF_{HH,pts\$} - \Delta MSF_{LL,pts\$}$ | $\boldsymbol{-\beta_1+\beta_2+\beta_3+\beta_4+\beta_6+\beta_7} = (\beta_0+\beta_2+\beta_3+\beta_4+\beta_6+\beta_7) - (\beta_0+\beta_1)$ |
| Advantageous own-pts | $\Delta MSF_{HH,pts} - \Delta MSF_{LL,pts}$ | $\boldsymbol{\beta_3+\beta_4+\beta_7} = (\beta_0+\beta_3+\beta_4+\beta_7) - \beta_0$ |
| Advantageous own-$ | $(\Delta MSF_{HH,pts\$} - \Delta MSF_{LL,pts\$}) - (\Delta MSF_{HH,pts} - \Delta MSF_{LL,pts})$ | $\boldsymbol{-\beta_1+\beta_2+\beta_6} = (-\beta_1+\beta_2+\beta_3+\beta_4+\beta_6+\beta_7) - (\beta_3+\beta_4+\beta_7)$ |

Bold indicates the simplified form of the non-bold formula to the right

In Panel A of Table 2, we provide a complete list of the relative- and own-rewards shocks and, in Panel B, the formulae we use to identify their impacts.[10]

### 3.4 Experimental predictions

#### 3.4.1 Relative-rewards-shocks predictions

Making no further assumptions, the RIE yields two unambiguous predictions about the impact of relative-rewards shocks:

- Positive impact of the advantageous relative-pts-$ shock:
  - $\Delta MSF_{HL,pts\$} - \Delta MSF_{HH,pts\$} > 0$
- Negative impact of the disadvantageous relative-pts-$ shock:
  - $\Delta MSF_{LH,pts\$} - \Delta MSF_{LL,pts\$} < 0$

These predictions can be inferred from the definition of the RIE, as both comparisons hold constant the subject's monetized pts and vary Participant X's, with smaller (larger)—in comparison to an equal—number of monetized pts for Participant X associated with higher (lower) values of ΔMSF.

Predicting the impact of the other relative-rewards shocks requires assumptions about MSF-scores' responsiveness to non-monetized pts. We will consider two alternative assumptions: (a) *strict RIE*, whereby the RIE applies to monetized but not non-monetized pts, versus (b) *generalized relative effect (RE)*, whereby the RIE applies to monetized pts and an analogous relative effect—such that SWB is decreasing in others' pts, holding own pts constant—applies with equal magnitude to non-monetized pts. These assumptions represent two extreme and non-exhaustive cases—an intermediate case could be that non-monetized pts impact mood but not as much as do monetized pts.

Under the strict RIE, the impacts of both advantageous and disadvantageous relative-pts shocks are predicted to be zero, and this yields the prediction that $-shocks will have the same impact as their corresponding pts-$-shocks. Specifically:

- Zero impact of the advantageous relative-pts shock:
  - $\Delta MSF_{HL,pts} - \Delta MSF_{HH,pts} = 0$
- Zero impact of the disadvantageous relative-pts shock:
  - $\Delta MSF_{LH,pts} - \Delta MSF_{LL,pts} = 0$
- Equal impacts of the advantageous relative-$ and advantageous relative-pts-$ shocks:

[10] The impact of disadvantageous own-rewards shocks are not listed, as they can be calculated by multiplying the impact of the corresponding advantageous own-rewards shocks by negative one.

  - $\Delta MSF_{HL,\$} - \Delta MSF_{HH,\$} = \Delta MSF_{HL,pts\$} - \Delta MSF_{HH,pts\$}$
- Equal impacts of the disadvantageous relative-$ and disadvantageous relative-pts-$ shocks:
  - $\Delta MSF_{LH,\$} - \Delta MSF_{LL,\$} = \Delta MSF_{LH,pts\$} - \Delta MSF_{LL,pts\$}$

Under generalized RE, the impacts of the advantageous and disadvantageous relative-pts shocks are predicted to equal the impacts of the corresponding pts-$ shocks, and this yields the prediction that the relative-$ shocks will have no impact.

- Equal impacts of the advantageous relative-pts and advantageous relative-pts-$ shocks:
  - $\Delta MSF_{HL,pts} - \Delta MSF_{HH,pts} = \Delta MSF_{HL,pts\$} - \Delta MSF_{HH,pts\$}$
- Equal impacts of the disadvantageous relative-pts and disadvantageous relative-pts-$ shocks:
  - $\Delta MSF_{LH,pts} - \Delta MSF_{LL,pts} = \Delta MSF_{LH,pts\$} - \Delta MSF_{LL,pts\$}$
- Zero impact of the advantageous relative-$ shock:
  - $\Delta MSF_{HL,\$} - \Delta MSF_{HH,\$} = 0$
- Zero impact of the disadvantageous relative-$ shock:
  - $\Delta MSF_{LH,\$} - \Delta MSF_{LL,\$} = 0$

### 3.4.2 Own-rewards-shock predictions

Assuming that subjects are made happier by receiving more rather than fewer monetized pts, and/or that, relative to expectations, subjects are made happier by gains of monetized pts in comparison to losses, the following prediction obtains:

- Positive impact of the advantageous own-pts-$ shock:
  - $\Delta MSF_{HH,pts\$} - \Delta MSF_{LL,pts\$} > 0$

Additional predictions about own-rewards shocks require additional assumptions. If we assume that, analogous to the strict RIE, MSF-scores do not respond to own non-monetized pts, then an advantageous own-pts shock is predicted to have no impact, and an advantageous own-$ shock will have the same impact as an advantageous own-pts-$ shock:

- Zero impact of the advantageous own-pts shock:
  - $\Delta MSF_{HH,pts} - \Delta MSF_{LL,pts} = 0$
- Equal impacts of the advantageous own-$ and advantageous own-pts-$ shocks:

- $\Delta MSF_{HH,\$} - \Delta MSF_{LL,\$} = \Delta MSF_{HH,pts\$} - \Delta MSF_{LL,pts\$}$

If we assume that, analogous to the generalized RE, MSF-scores respond equally to own non-monetized and monetized pts, then an advantageous own-pts shock is predicted to have the same impact as an advantageous own-pts-$ shock, and an advantageous own-$ shock will have no impact:

- Equal impacts of the advantageous own-pts and advantageous own-pts-$ shocks:
  - $\Delta MSF_{HH,pts} - \Delta MSF_{LL,pts} = \Delta MSF_{HH,pts\$} - \Delta MSF_{LL,pts\$}$
- Zero impact of the advantageous own-$ shock:

  $\Delta MSF_{HH,\$} - \Delta MSF_{LL,\$} = 0$

## 4 Results

Of the 996 subjects who completed the experiment, 471 were in the pts-treatment and 525 in the pts$-treatment. 132 subjects (69 in the pts-treatment and 63 in the pts$-treatment) completed at least one of the screening items incorrectly and were dropped from the analysis.[11] Table 3 presents the demographic characteristics of the sample, which are comparable to other mTurk samples in the literature (see Kuziemko et al. (2015) and Snowberg and Yariv (2018)). Table 4 presents the pre- and post-revelation MSF-scores and ΔMSF for each cell.

### 4.1 Non-parametric tests of own- and relative-rewards shocks

In Table 5, we present the estimated impacts on ΔMSF of all own- and relative-rewards shocks.[12] Because we use multiple outcomes, we use Bonferroni adjustments to control for the familywise error rate; List et al. (2016) consider this the most conservative approach. The Bonferroni adjustments are applied to alphas, not *p* values; so the *p* values reported in Table 5 are unadjusted. Bonferroni-adjusted alphas are equal to the standard Type 1 error rate (e.g., 0.05) divided by the number of tests in a family; starring in the table corresponds to whether the raw *p* value is less than the Bonferroni-adjusted alpha. We define three families, each of which has three tests. The families are: advantageous own-rewards (first three rows of Table 5),

[11] 2 subjects did not indicate that the year was "2017;" 9 subjects did not "Disagree" or "Strongly Disagree" that the current year was 2025; 53 (45) subjects indicated the incorrect payment for themselves (Participant X) in the pts$-treatment; and 56 (64) subjects indicated the incorrect payment for themselves (Participant X) in the pts-treatment. The main results, using the methods presented in Sect. 4.2 below, hold if we do not drop these subjects (see columns (3) and (4) of Appendix Table 7). It can be argued that keeping these subjects in the analysis may be preferred, as it may be more reflective of a general population in which some individuals do not pay close attention to their own and/or others' income; thank you to Alan Kirman for pointing this out.

[12] Columns (5) and (6) of Appendix Table 7 show, using the methods from Sect. 4.2 below, that the relative- and own-rewards shocks did not impact BPNSS scores.

**Table 3** Demographic characteristics, n = 860

| | Mean |
|---|---|
| *Gender* | |
| Female | 0.45 |
| Male | 0.54 |
| Other / prefer not to answer | 0.01 |
| Age | 36.48 |
| *Are you a citizen or permanent resident of the United States?* | |
| Yes | 1.00 |
| No | 0.00 |
| Prefer not to answer | 0.00 |
| *Rate your English* | |
| Native | 0.98 |
| Fluent | 0.02 |
| Proficient | 0.00 |
| *What race/ethnicity do you identify yourself as?* | |
| American Indian and Alaska Native | 0.00 |
| Asian | 0.08 |
| Black or African | 0.07 |
| Hispanic | 0.05 |
| White | 0.77 |
| Other / prefer not to answer | 0.02 |
| *What religion do you consider yourself?* | |
| Atheist / agnostic | 0.47 |
| Buddhist | 0.02 |
| Christian | 0.42 |
| Hindu | 0.00 |
| Jewish | 0.02 |
| Muslim | 0.00 |
| Other / prefer not to answer | 0.07 |
| *How would you characterize your political views?* | |
| Conservative | 0.20 |
| Liberal / progressive | 0.50 |
| Moderate | 0.28 |
| Prefer not to answer | 0.02 |
| *What is the total (gross) income last year of your household?* | |
| $0–25,000 | 0.18 |
| $25,000–50,000 | 0.30 |
| $50,000–75,000 | 0.24 |
| $75,000–100,000 | 0.15 |
| $100,000–125,000 | 0.05 |
| $125,000–150,000 | 0.03 |
| $150,000 + | 0.04 |
| Prefer not to answer | 0.02 |

**Table 3** (continued)

| | Mean |
|---|---|
| *What is your highest level of education?* | |
| Some high school | 0.00 |
| Completed high school | 0.10 |
| Some college | 0.28 |
| Completed college | 0.44 |
| Some grad/professional school | 0.04 |
| Completed grad/professional school | 0.14 |
| Prefer not to answer | 0.00 |
| *Please indicate your employment status* | |
| Employed, full-time | 0.67 |
| Employed, part-time | 0.15 |
| Not employed, looking for work | 0.06 |
| Not employed, not looking for work | 0.06 |
| Retired | 0.02 |
| Student | 0.03 |
| Prefer not to answer | 0.02 |

advantageous relative-rewards (next three rows of Table 5), and disadvantageous relative-rewards (last three rows of Table 5).

Considering the own-rewards shocks, we observe that the impact of advantageous own-pts-$, own-pts, and own-$ shocks are positive, statistically significant, and economically meaningful. For example, the impact of the advantageous own-$ shock is 2.17 MSF-points. In other words, controlling for the impact of own pts, subjects who receive $10 rather than $2 *ceteris paribus* experience a mood-improvement of roughly 14% of the pre-revelation sample-mean MSF-score (=2.17/15.25). The impact of the advantageous own-pts shock being positive and significant indicates that subjects' mood is impacted by receiving non-monetized pts. The impact of the advantageous own-$ shock being positive and significant indicates that the impact on subjects' mood of monetized pts exceeds the impact of receiving equivalent non-monetized pts.

Turning to the advantageous relative-rewards shocks, there is support for neither the strict RIE, nor the generalized RE. Table 5 shows that the impacts of the advantageous relative-pts-$ and relative-pts shocks are both negative, with the latter statistically significantly so. While the impact of the advantageous relative-$ shock is positive—consistent with the strict RIE insofar as it is less mood-diminishing to be "richer" than other subjects in both pts and $ than in pts alone—it is not statistically significant. Taken together, there is no evidence that it is mood-improving to be "richer" than other subjects. If anything, the results of the advantageous relative-pts shock are suggestive of it being mood-diminishing. We can only speculate as to why this is unique to advantageous relative-pts and not advantageous relative-pts-$ shocks. One possibility could be that subjects

**Table 4** Pre- and post-revelation MSF-scores and ΔMSF by cell

| Cell | MSF-score | pts-$-Treatment | pts-Treatment |
|---|---|---|---|
| LL | Pre-revelation | 15.081 | 15.495 |
| | | (3.451) | (2.950) |
| | Post-revelation | 13.970 | 15.495 |
| | | (3.947) | (3.340) |
| | Change | − 1.111 | 0.000 |
| | | (2.740) | (1.788) |
| | | *p = 0.0009* | *p = 0.9417* |
| | Observations | 99 | 103 |
| LH | Pre-revelation | 14.922 | 15.055 |
| | | (3.389) | (3.410) |
| | Post-revelation | 12.767 | 14.312 |
| | | (4.225) | (3.736) |
| | Change | − 2.155 | − 0.743 |
| | | (3.247) | (2.605) |
| | | *p = 0.0000* | *p = 0.0036* |
| | Observations | 103 | 109 |
| HL | Pre-revelation | 15.398 | 16.056 |
| | | (3.109) | (2.936) |
| | Post-revelation | 17.293 | 16.539 |
| | | (2.983) | (2.735) |
| | Change | 1.894 | 0.483 |
| | | (2.472) | (1.439) |
| | | *p = 0.0000* | *p = 0.0051* |
| | Observations | 123 | 89 |
| HH | Pre-revelation | 15.030 | 15.139 |
| | | (3.481) | (3.516) |
| | Post-revelation | 17.286 | 16.337 |
| | | (2.933) | (3.226) |
| | Change | 2.256 | 1.198 |
| | | (2.338) | (1.709) |
| | | *p = 0.0000* | *p = 0.0000* |
| | Observations | 133 | 101 |

Standard deviations are in parentheses. *p* values are from Wilcoxon signed-rank tests of the equality of the distributions from which pre- and post-revelation MSF-scores are drawn

have social-surplus-maximization concerns over monetized but not non-monetized rewards, leading to mood-improvement for subjects in $HH_{pts\$}$ (in which total social surplus is $20) relative to subjects in $HL_{pts\$}$ (in which total social surplus is $12). Note that this would not be the case in $HH_{pts}$ relative to $HL_{pts}$, as the total social surplus is $12 in both. In other words, in both the advantageous relative-pts

**Table 5** Estimated impacts of relative- and own-rewards shocks on ΔMSF

| Shocks | Formulae | |
|---|---|---|
| Adv own-pts-$ | $\Delta MSF_{HH,pts\$} - \Delta MSF_{LL,pts\$}$ | 3.367*** |
| | | (0.342) |
| | | $p = 0.0000$ |
| Adv own-pts | $\Delta MSF_{HH,pts} - \Delta MSF_{LL,pts}$ | 1.198*** |
| | | (0.245) |
| | | $p = 0.0000$ |
| Adv own-$ | $(\Delta MSF_{HH,pts\$} - \Delta MSF_{LL,pts\$}) - (\Delta MSF_{HH,pts} - \Delta MSF_{LL,pts})$ | 2.169*** |
| | | (0.421) |
| | | $p = 0.0000$ |
| Adv rel-pts-$ | $\Delta MSF_{HL,pts\$} - \Delta MSF_{HH,pts\$}$ | −0.361 |
| | | (0.302) |
| | | $p = 0.3256$ |
| Adv rel-pts | $\Delta MSF_{HL,pts} - \Delta MSF_{HH,pts}$ | −0.715*** |
| | | (0.228) |
| | | $p = 0.0021$ |
| Adv rel-$ | $(\Delta MSF_{HL,pts\$} - \Delta MSF_{HH,pts\$}) - (\Delta MSF_{HL,pts} - \Delta MSF_{HH,pts})$ | 0.354 |
| | | (0.378) |
| | | $p = 0.5775$ |
| Disadv rel-pts-$ | $\Delta MSF_{LH,pts\$} - \Delta MSF_{LL,pts\$}$ | -1.044** |
| | | (0.422) |
| | | $p = 0.0095$ |
| Disadv rel-pts | $\Delta MSF_{LH,pts} - \Delta MSF_{LL,pts}$ | −0.743* |
| | | (0.305) |
| | | $p = 0.0208$ |
| Disadv rel-$ | $(\Delta MSF_{LH,pts\$} - \Delta MSF_{LL,pts\$}) - (\Delta MSF_{LH,pts} - \Delta MSF_{LL,pts})$ | −0.301 |
| | | (0.521) |
| | | $p = 0.5618$ |

Standard deviations are in parentheses. $p$ values are from Mann–Whitney tests; for example, the first row tests the equality of distributions from which $\Delta MSF_{HH,pts\$}$ and $\Delta MSF_{LL,pts\$}$ are drawn. Bonferroni adjusted alphas are used for starring so *, **, *** represent $p$ values < 0.0333, 0.0167, and 0.0033, respectively (representing standard Type 1 error rates of 0.10, 0.05, and 0.01 divided by the number of tests in a family; the advantageous-own-rewards family has three tests (first three rows); the advantageous-relative-rewards family has three tests (next three rows); and the disadvantageous-relative-rewards family has three tests (last three rows)

and advantageous relative-pts-$ treatments, it may be mood-diminishing to be "richer" in points than Participant X, but in the latter treatment, this is offset by mood-improvements due to social surplus maximization of monetized rewards.[13]

[13] It should be noted that the results for disadvantageous relative-rewards shocks presented below are not consistent with an analogous concern for social surplus maximization, suggesting either that such concerns are not driving the advantageous relative-reward results or that social surplus concerns are less effective at tempering mood in the LLpts$ and LHpts$ conditions (in which subjects learn their payments are lower than expected) than in HHpts$ and HLpts$ (in which subjects learn their payments are higher than expected).

The estimated impacts of the disadvantageous relative-rewards shocks, though, favor the generalized RE. The impacts of the relative-pts-$ and relative-pts shocks are negative and significant. Further, they are statistically indistinguishable from each other, yielding an insignificant impact of the disadvantageous relative-$ shock. This suggests that being "poorer" than other subjects in monetized points is only as mood-diminishing as is being "poorer" in non-monetized points.

Comparing the impacts of the own-rewards and relative-rewards shocks reveals two interesting asymmetries. First, one's mood seems to depend on her own money—but not others' money—more than the points the money represents. Second, one's mood diminishes from being behind others but does not improve from being ahead.

### 4.2 Regression and subgroup analyses

To analyze the MSF-effects of relative-rewards shocks while controlling for demographic characteristics, we estimate the following equation:

$$\Delta\text{MSF} = \beta_0 + \beta_1 \text{I}_{2\text{pts\$}} + \beta_2 \text{I}_{10\text{pts\$}} + \beta_3 \text{I}_{10\text{pts}} + \beta_4 \text{I}_{\text{X10pts}} + \beta_5 \text{I}_{2\text{pts\$,X10pts\$}} + \beta_6 \text{I}_{10\text{pts\$,X10pts\$}} + \beta_7 \text{I}_{10\text{pts,X10pts}} + \delta \text{Y} + \varepsilon \tag{1}$$

The indicator variable $\text{I}_{2\text{pts\$}}$ ($\text{I}_{10\text{pts\$}}$) equals one if the subject receives 2 pts (10 pts) worth $1 each. The indicator variable $\text{I}_{10\text{pts}}$ ($\text{I}_{\text{X10pts}}$) equals one if the subject (Participant X) receives 10 pts. The indicator variable $\text{I}_{2\text{pts\$,X10pts\$}}$ ($\text{I}_{10\text{pts\$,X10pts\$}}$) equals one if the subject receives 2 pts (10 pts) worth $1 each and Participant X receives 10 pts worth $1 each. The indicator variable $\text{I}_{10\text{pts,X10pts}}$ equals one if the subject receives 10 pts and Participant X receives 10 pts. The vector Y contains demographic characteristics.[14] Robust standard errors are calculated.

The constant term $\beta_0$ captures the MSF-change experienced by subjects in the $\text{LL}_{\text{pts}}$ cell. Regression coefficients can be used to recover the impacts of relative- and own-rewards shocks; the formulae are presented in Panel C of Table 2 and the corresponding estimates are reported in Table 6. Column (1) of Table 6 excludes controls for demographic characteristics and replicates the corresponding results in Table 5. Column (2) includes controls for demographic characteristics, yielding results similar to column (1).

[14] Demographic characteristics—all of which are treated as categorical variables in the regression analysis—include gender (female, male, other, prefer not to answer), age (bottom-quartile (< 29 years) and top-quartile (> 42 years)), race (American Indian/Alaska Native, Asian, Black/African American, Hispanic, Native Hawaiian/other Pacific Islander, White, other, prefer not to answer), religion (Atheist/Agnostic, Buddhist, Christian, Hindu, Jewish, Muslim, other, prefer not to answer), political leaning (conservative, liberal/progressive, moderate, prefer not to answer), household income ($0-$25 K, $25 K-$50 K, $50 K-$75 K, $75 K-$100 K, $100 K-$125 K, $125 K-$150 K, $150 K+, prefer not to answer), education (some high school, completed high school, some college, completed college, some grad/professional school, completed grad/professional school, prefer not to answer), and employment status (employed, full-time; employed, part-time; not employed, looking for work; not employed, not looking for work; retired; student; prefer not to answer).

**Table 6** Regression estimates of the impacts of relative- and own-rewards shocks on ΔMSF

| Shocks | All | All | Conservative | Liberal / Progressive | Female | Male | Income in top quartile | Income in bottom 3 quartiles |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| *Panel A: Impacts of advantageous relative-rewards shocks* | | | | | | | | |
| Adv rel-pts-$ | −0.361 | −0.431 | −0.780 | −0.554 | −0.991* | −0.076 | −0.369 | −0.311 |
| | (0.302) | (0.313) | (0.671) | (0.496) | (0.490) | (0.449) | (0.673) | (0.346) |
| | $p=0.231$ | $p=0.170$ | $p=0.247$ | $p=0.265$ | $p=0.044$ | $p=0.866$ | $p=0.584$ | $p=0.369$ |
| | | | $p$ value of equality = 0.787 | | $p$ value of equality = 0.169 | | $p$ value of equality = 0.939 | |
| Adv rel-pts | −0.715*** | −0.800*** | −0.953 | −0.463 | −1.435*** | −0.296 | −0.776 | −0.855*** |
| | (0.228) | (0.242) | (0.553) | (0.339) | (0.437) | (0.295) | (0.631) | (0.285) |
| | $p=0.002$ | $p=0.001$ | $p=0.087$ | $p=0.173$ | $p=0.001$ | $p=0.316$ | $p=0.221$ | $p=0.003$ |
| | | | $p$ value of equality = 0.450 | | $p$ value of equality = 0.031 | | $p$ value of equality = 0.909 | |
| Adv rel-$ | 0.354 | 0.370 | 0.173 | −0.091 | 0.445 | 0.220 | 0.406 | 0.544 |
| | (0.378) | (0.395) | (0.841) | (0.602) | (0.659) | (0.549) | (0.943) | (0.448) |
| | $p=0.350$ | $p=0.349$ | $p=0.837$ | $p=0.880$ | $p=0.500$ | $p=0.689$ | $p=0.667$ | $p=0.225$ |
| | | | $p$ value of equality = 0.798 | | $p$ value of equality = 0.793 | | $p$ value of equality = 0.895 | |
| *Panel B: Impacts of disadvantageous relative-rewards shocks* | | | | | | | | |
| Disadv rel-pts-$ | −1.044** | −1.026** | −2.620* | −0.106 | −1.748** | −0.293 | −1.975* | −0.788 |
| | (0.422) | (0.432) | (1.176) | (0.590) | (0.664) | (0.598) | (0.925) | (0.503) |
| | $p=0.014$ | $p=0.018$ | $p=0.028$ | $p=0.858$ | $p=0.009$ | $p=0.624$ | $p=0.034$ | $p=0.118$ |
| | | | $p$ value of equality = 0.056 | | $p$ value of equality = 0.104 | | $p$ value of equality = 0.260 | |
| Disadv rel-pts | −0.743** | −0.665** | −0.518 | −0.714 | −0.686 | −0.500 | −0.708 | −0.520 |
| | (0.305) | (0.318) | (0.538) | (0.405) | (0.511) | (0.413) | (0.607) | (0.399) |
| | $p=0.015$ | $p=0.037$ | $p=0.337$ | $p=0.078$ | $p=0.180$ | $p=0.227$ | $p=0.245$ | $p=0.193$ |
| | | | $p$ value of equality = 0.772 | | $p$ value of equality = 0.777 | | $p$ value of equality = 0.795 | |

**Table 6** (continued)

| Shocks | All | All | Conservative | Liberal / Progressive | Female | Male | Income in top quartile | Income in bottom 3 quartiles |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Disadv rel-$ | −0.301 | −0.361 | −2.101 | 0.608 | −1.061 | 0.207 | −1.267 | −0.268 |
| | (0.521) | (0.537) | (1.294) | (0.718) | (0.822) | (0.702) | (1.120) | (0.647) |
| | $p=0.563$ | $p=0.501$ | $p=0.107$ | $p=0.397$ | $p=0.198$ | $p=0.769$ | $p=0.260$ | $p=0.679$ |
| | | | $p$ value of equality = 0.067 | | $p$ value of equality = 0.241 | | $p$ value of equality = 0.440 | |
| *Panel C: Impacts of advantageous own-rewards shocks* | | | | | | | | |
| Adv own-pts-$ | 3.367*** | 3.392*** | 2.934*** | 4.136*** | 3.136*** | 3.625*** | 2.820*** | 3.548*** |
| | (0.342) | (0.353) | (0.660) | (0.562) | (0.484) | (0.542) | (0.711) | (0.415) |
| | $p=0.000$ | $p=0.000$ | $p=0.000$ | $p=0.000$ | $p=0.000$ | $p=0.000$ | $p=0.000$ | $p=0.000$ |
| | | | $p$ value of equality = 0.166 | | $p$ value of equality = 0.502 | | $p$ value of equality = 0.376 | |
| Adv own-pts | 1.198*** | 1.294*** | 1.208** | 1.167*** | 1.901*** | 0.915** | 0.898 | 1.369*** |
| | (0.245) | (0.256) | (0.508) | (0.350) | (0.445) | (0.337) | (0.543) | (0.327) |
| | $p=0.000$ | $p=0.000$ | $p=0.019$ | $p=0.001$ | $p=0.000$ | $p=0.007$ | $p=0.100$ | $p=0.000$ |
| | | | $p$ value of equality = 0.947 | | $p$ value of equality = 0.078 | | $p$ value of equality = 0.457 | |
| Adv own-$ | 2.169*** | 2.098*** | 1.726* | 2.969*** | 1.236 | 2.710*** | 1.922* | 2.179*** |
| | (0.421) | (0.433) | (0.845) | (0.643) | (0.650) | (0.637) | (0.886) | (0.529) |
| | $p=0.000$ | $p=0.000$ | $p=0.043$ | $p=0.000$ | $p=0.058$ | $p=0.000$ | $p=0.031$ | $p=0.000$ |
| | | | $p$ value of equality = 0.242 | | $p$ value of equality = 0.105 | | $p$ value of equality = 0.803 | |
| Controls included | No | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| R squared | 0.278 | 0.311 | 0.477 | 0.347 | 0.360 | 0.326 | 0.423 | 0.310 |
| Observations | 860 | 860 | 175 | 434 | 391 | 463 | 200 | 646 |

Robust standard errors are reported in parentheses. In columns (1)&(2), *, **, *** indicate $p$ values < 0.10, 0.05, and 0.01, respectively. Bonferroni adjusted alphas are used for starring in columns (3)–(8) to account for the two subgroups in columns (3) and (4), (5) and (6), and (7) and (8), respectively; so in those columns *, **, *** indicate $p$ values < 0.05, 0.025, and 0.005. $p$ values from tests of equality of coefficients across columns (3) and (4), (5) and (6), and (7) and (8) are presented below the corresponding coefficients

Turning to subgroup analyses, we investigate all differences that are suggested by the SWB and social preferences literature for which we have demographic data in the questionnaire. Alesina et al. (2004) find that conservatives' SWB is unaffected by—while progressives' SWB decreases with—income inequality in the US; this may be due to conservatives being more likely to attribute success to hard work and talent, and progressives to luck (Frank 2016). Kamas and Preston (2015) identify gender-differences in social preferences, with women more likely to be inequity averse and men more likely to be social-surplus maximizers. Ifcher et al. (2018) and Alesina et al. (2004) find that the negative income-inequality-SWB relationship is significantly stronger for high- than low-income Americans, with no significant relationship for low-income subgroups in some specifications. Kuziemko et al. (2014) find that "last-place aversion" is particularly strong for conservatives and individuals with low income. Subgroup analyses by political orientation, gender, and income are presented in columns (3)–(8) of Table 6.[15]

The most notable difference by political orientation is that the disadvantageous relative-pts-$ shock is significantly mood-diminishing for conservatives and has no significant mood-effect for liberal/progressives. Further, the mood-effects for conservatives and liberal/progressives are marginally significantly different (−2.260 vs. −0.106, $p$ value = 0.056).[16] In sum, an interesting asymmetry by political orientation is suggested: being "richer" may not affect conservatives and liberal/progressives differently, but being "poorer" may. This would be consistent with Kuziemko et al. (2014) and may be explained by conservatives' strong last-place aversion.

Subgroup analyses by gender reveal that the impact of the advantageous relative-pts-$, advantageous relative-pts, and disadvantageous relative-pts-$ shocks are significantly mood-diminishing for women but not men. Further, the mood-effect of the advantageous relative-pts shock is significantly greater in magnitude for women than men (−1.435 vs. −0.296, $p$ value = 0.031). In sum, there is evidence that both being "richer" and being "poorer" are mood-diminishing for women but neither has mood-effects for men. This finding is consistent with Kamas and Preston (2015) and may be explained by women being more inequity-averse than men. There is also weak evidence of gender differences in the mood-effects of own-pts versus own-$ shocks: while the impact of the advantageous own-pts-$ shock is similar for men and women, the impact of the advantageous own-pts shock is marginally significantly more mood-improving for women than men (1.901 vs. 0.915, $p$ value = 0.078).

[15] Because we divide the sample into subgroups, we use Bonferroni adjustments in these columns. As in Table 5, Bonferroni adjustments are applied to alphas, not p-values; so the p-values reported in Table 6 are unadjusted, and starring in the table corresponds to whether the raw p-value is less than the Bonferroni-adjusted alpha. Bonferroni-adjusted alphas are equal to the standard Type 1 error rate (e.g., 0.05) divided by two, the number of mutually exclusive subgroups the sample is divided into (e.g., male and female).

[16] Because the mood-effects of the disadvantageous relative-pts shocks are similar by political orientation, this means the mood-effects of the disadvantageous relative-$ shocks are different for conservatives and liberal/progressives, though not statistically significantly so at conventional levels (−2.101 vs. 0.608, $p$ value = 0.107).

Lastly, the only notable difference by income-subgroups is that the disadvantageous relative-pts-$ shock is significantly mood-diminishing for individuals with household income in the top-quartile and not for those in the bottom 3 quartiles.[17] This finding is consistent with Ifcher et al. (2018) and Alesina et al. (2004), but not Kuziemko et al. (2014).

## 5 Discussion

In an experiment designed to test for the RIE, we find no support for an interpretation that applies strictly to income; this is because we find that the impact on mood of being "richer" or "poorer" in experimental points is the same when points are worth money and when they are not. This cannot be explained by money and points impacting mood equally, as receiving more money than expected improves subjects' moods by significantly more than does receiving equivalently more points than expected. Further, we find only partial support for an interpretation of a generalized RE that applies to advantageous and disadvantageous relative-rewards shocks alike; this is because we find evidence of a worsening of mood from being "poorer" than others, but no evidence of an improvement in mood from being "richer" than others.

Our attempt to disentangle the RIE's income from non-income relativity concerns is novel. Further, we do so with an extreme counterfactual: non-income relativity is manipulated using a non-monetized, cardinal point system.

What are the implications of our study for research that tests for the RIE using SWB and income data from surveys? As noted in the literature review, in such studies, the SWB-impact of non-relative factors like cost-of-living may be misattributed to others' income in the absence of proper controls. Our findings suggest that, even if a significant negative coefficient on others' income obtains with the inclusion of controls for non-relative factors like cost-of-living, the negative coefficient may not be attributable to concerns about how one's own income compares to others', but instead to other social-comparison concerns. Indeed, people may be biased toward using income because it is often "observed" as a number (e.g., a statistic) or as a material good (e.g., by proxy as conspicuous consumption), and because numbers and material income may be used as a proxy for ineffable factors in social comparisons that, if defined and concretized, would weaken relative income's relationship to SWB.

For example, suppose SWB's only determinant is feeling like a productive member of society—a latent variable for researchers.[18] Further suppose that though the

[17] The advantageous relative-pts shock does not significantly impact the mood of individuals with household income in the top quartile, but it is significantly mood-diminishing for those in the bottom 3 quartiles. While this may seem like an important difference, the similar magnitudes of the point-estimates ($-0.776$ vs. $-0.855$, $p$ value $=0.909$) suggest that it is likely due to variant sample sizes ($n=200$ versus 646).

[18] The example works just as well if we substitute "feeling like a productive member of society" with a possibly even more ineffable factor like "feeling high status".

feeling itself is observable to the individual having it, it is determined by a latent process that takes as inputs signals of others' productivity, own productivity, and socially held standards of productivity. Four cases emerge, depending on the availability of more direct productivity information: (1) If information on productivity is unavailable to both researchers and individuals, then the individual will use observed conspicuous consumption to infer others' income to, in turn, infer others' and own productivity (and/or use reported income data to infer productivity). In this case, RIE researchers will attribute traditional regression results to RIE. (2) If information on productivity is available to researchers but not individuals, individuals will continue to use conspicuous consumption and/or reported income data to infer productivity. Controlling for productivity will diminish the coefficient on others' income. If the coefficient is not nullified, the RIE will be supported, and the role of productivity ruled out. (3) If information on productivity is available to individuals but not researchers, individuals will cease to use signals to infer productivity. If because of correlation between productivity and the input signals, the coefficient on others' income is still significant, researchers will attribute it to the RIE. (4) Only in the case that information on productivity is available to both researchers and individuals will regression analysis rule out the RIE and properly attribute SWB effects to productivity. Prior evidence regarding the RIE implicitly measures the impact of something akin to our relative-pts-$, rather than relative-$, shocks. The implication for RIE researchers depends on whether they are interested in distinguishing between relative income as a determinant of SWB and relative income as a signal-of-convenience for other determinants of SWB.[19]

In our experiment, of course, there is no scope for productivity, and subjects are informed, depending on treatment, either of a perfect correlation between points and money or a zero correlation between points and money. That we observe relative-pts shocks affecting mood suggests that numerical information, innately hierarchical as it is, is particularly salient in the latent process by which social information affects feelings. While points are explicitly non-monetized in our points treatment, their usual connotation as a reward together with the ease with which they, as numbers, convey relative standing may be enough to generate their mood-effects. This may have the same perceptual roots as money illusion, the observed tendency to understand prices in nominal rather than real terms. In related work, Butler (2016) conducts a laboratory experiment in which subjects are randomized into receiving high or low wages for the same task; they then perform the task and report their beliefs about their performance. Despite objectively

[19] Other experimental evidence of money's protean nature as a signal comes from the experimental psychology literature. Priming the concept of money (by, for example, showing subjects images of money or having them handle paper currency) increases subjects' self-sufficiency (Vohs et al. 2006), tolerance for physical pain (Zhou et al. 2009), work and productivity (Mogilner 2010); and decreases the savoring of experiences (Quoidbach et al. 2010), willingness to volunteer or donate (Pfeffer and DeVoe 2009; Chatterjee et al. 2013), socializing and happiness (Mogilner 2010), and anxiety and fear of death (Zaleskiewicz et al. 2013).

equal task-performance and subjects' knowing their wages have been randomized, those assigned to high wages believe their performance to be significantly better than those assigned to low wages.

Our research also has possible implications for policy makers, particularly those for whom citizens' SWB is an outcome of interest. First let's consider the policy implications of the RIE. The RIE poses a problem for policy makers seeking to increase SWB by pursuing policies that increase income across the distribution: the SWB gains from increases to individuals' own income would be at least partially offset by SWB losses from increases to others' income. The RIE does not advance the neoclassical economic understanding of the optimal mean-preserving distribution of income: as the variance of income increases, any SWB gains for those with higher-than-average income may be offset by SWB losses for those with lower-than-average income. Our results suggest zero SWB gains for those with higher-than-average income and SWB losses for those with lower-than-average income; this means that, *ceteris paribus*, more compressed and perhaps left-skewed income distributions would be favored.

Our approach, of course, begs the question of whether SWB-change is predictive of choice, and whether a revealed-preference approach would result in the same patterns we observe. In Benjamin et al. (2014a), graduating medical students' actual preference-rankings of residency programs are compared to their reported beliefs about various features of each program and their predictions of various aspects of their own experience (including their SWB) during and after each program. Predicted SWB is found to be the best, but not the only important, predictor of actual preference-rankings. While this study sheds some light on the relationship between SWB and revealed-preference, SWB is only considered prospectively and not contemporaneously. Further research is necessary to directly address whether our mood-change approach is compatible with a revealed-preference approach. If so, it could substitute for revealed-preference in other contexts, like identifying preferences for earned versus unearned income, or estimating parameters related to loss aversion, time-, risk-, or competitive-preferences. If not, it could potentially complement revealed-preference. For example, it may be that our approach, being emotionally based, reflects System-1 thinking while revealed-preference, being deliberative, reflects System-2 thinking, and that preferences may be best understood using a combination of the two approaches (Kahneman 2011).

**Acknowledgements** Financial support from Barnard College is gratefully acknowledged.

## Appendix

See Table 7.

**Table 7** Regression estimates of the impacts of relative- and own-rewards shocks on ΔMSF when including dropped subjects and when including pilot-study subjects, and on post-revelation 10-item BPNSS score

| Shocks | Include subjects from pilot study | Include subjects from pilot study | Include dropped subjects | Include dropped subjects | Post-revelation BPNSS score (10 items) | Post-revelation BPNSS score (10 items) |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A: Impacts of advantageous relative-rewards shocks* | | | | | | |
| Adv rel-pts-$ | −0.400 | −0.429 | −0.118 | −0.118 | 0.092 | 0.114 |
| | (0.269) | (0.313) | (0.278) | (0.284) | (0.124) | (0.127) |
| Adv rel-pts | −0.595*** | −0.801*** | −0.622*** | −0.677*** | 0.115 | 0.118 |
| | (0.200) | (0.242) | (0.235) | (0.251) | (0.131) | (0.124) |
| Adv rel-$ | 0.195 | 0.371 | 0.504 | 0.559 | −0.023 | −0.004 |
| | (0.335) | (0.395) | (0.364) | (0.375) | (0.180) | (0.176) |
| *Panel B: Impacts of disadvantageous relative-rewards shocks* | | | | | | |
| Disadv rel-pts-$ | −1.019*** | −1.026** | −0.990** | −0.939** | −0.152 | −0.186 |
| | (0.388) | (0.432) | (0.396) | (0.400) | (0.158) | (0.154) |
| Disadv rel-pts | −0.497* | −0.663** | −0.631** | −0.484 | −0.066 | −0.085 |
| | (0.256) | (0.318) | (0.291) | (0.300) | (0.148) | (0.148) |
| Disadv rel-$ | −0.522 | −0.363 | −0.359 | −0.454 | −0.087 | −0.101 |
| | (0.465) | (0.537) | (0.491) | (0.500) | (0.216) | (0.212) |
| *Panel C: Impacts of advantageous own-rewards shocks* | | | | | | |
| Adv own-pts-$ | 3.187*** | 3.392*** | 3.100*** | 3.137*** | −0.015 | −0.048 |
| | (0.315) | (0.353) | (0.314) | (0.321) | (0.149) | (0.148) |
| Adv own-pts | 1.104*** | 1.297*** | 1.285*** | 1.366*** | 0.182 | 0.201 |
| | (0.213) | (0.257) | (0.236) | (0.248) | (0.141) | (0.138) |
| Adv own-$ | 2.083*** | 2.096*** | 1.814*** | 1.771*** | −0.197 | −0.248 |
| | (0.380) | (0.434) | (0.393) | (0.400) | (0.205) | (0.202) |
| Controls included | No | Yes | No | Yes | No | Yes |

**Table 7** (continued)

| Shocks | Include subjects from pilot study | Include subjects from pilot study | Include dropped subjects | Include dropped subjects | Post-revelation BPNSS score (10 items) | Post-revelation BPNSS score (10 items) |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Observations | 1054 | 1054 | 996 | 996 | 860 | 860 |

Robust standard errors are reported in parentheses. *, **, *** indicate $p$ values < 0.10, 0.05, and 0.01, respectively. Controls include gender, age, race, religion, political leaning, household income, education, and employment status

## References


Alesina, A., Di Tella, R., & MacCulloch, R. (2004). Inequality and happiness: Are Europeans and Americans different? *Journal of Public Economics, 88*, 2009–2042.

Benjamin, D. J., Cooper, K., Heffetz, O., & Kimball, M. S. (2019). Self-reported wellbeing indicators are a valuable complement to traditional economic indicators but aren't yet ready to compete with them. *Behavioural Public Policy ,* forthcoming.

Benjamin, D. J., Heffetz, O., Kimball, M. S., & Rees-Jones, A. (2014a). Can marginal rates of substitution be inferred from happiness data? Evidence from residency choices". *American Economic Review, 104*(11), 3498–3528.

Benjamin, D. J., Heffetz, O., Kimball, M. S., & Rees-Jones, A. (2012). What do you think would make you happier? What do you think you would choose? *American Economic Review, 102*(5), 2083–2110.

Benjamin, D. J., Heffetz, O., Kimball, M. S., & Szembrot, N. (2014b). Beyond happiness and satisfaction: Toward well-being indices based on stated preference. *American Economic Review, 104*(9), 2698–2735.

Blanchflower, D. G., & Oswald, A. J. (2004). Well-being over time in Britain and the USA. *Journal of Public Economics, 88*(7), 1359–1386.

Bosman, R., & van Winden, F. (2002). Emotion hazard in a power-to-take experiment. *Economic Journal, 112*(476), 147–169.

Brodeur, A., & Fleche, S. (2019). Neighbors' income, public goods and well-being: Evidence from a multi-scale analysis. *Review of Income and Wealth, 65*(2), 217–238.

Butler, J. (2016). Inequality and relative ability beliefs. *Economic Journal, 26*, 907–948.

Charness, G., & Grosskopf, B. (2001). Relative payoffs and happiness: An experimental study. *Journal of Economic Behavior and Organization, 45*(3), 301–328.

Chatterjee, P., Rose, R. L., & Sinha, J. (2012). Why money meanings matter in decisions to donate time and money. *Marketing Letters, 24*, 1–10.

Clark, A. E., Frijters, P., & Shields, M. A. (2008). Relative income, happiness, and utility: An explanation for the Easterlin paradox and other puzzles. *Journal of Economic Literature, 46*(1), 95–144.

Clark, A. E., Kristensen, N., & Westergård-Nielsen, N. (2009a). Job satisfaction and co- worker wages: Status or signal? *Economic Journal, 119*, 430–447.

Clark, A. E., Kristensen, N., & Westergard-Neilsen, N. (2009b). Economic satisfaction and income rank in small neighbourhoods. *Journal of the European Economic Association, 7*(2–3), 519–527.

Clark, A. E. & Senik, C. (2011). Is happiness different from flourishing? Cross-country evidence from the ESS. In *Revue d'économie politique , numéro spécial "Les nouveaux indicateurs de bien-être", 1*, 17–34.

D'Ambrosio, C., & Frick, J. R. (2012). Individual well-being in a dynamic perspective. *Economica, 79*, 284–302.

Deaton, A., & Stone, A. A. (2013). Two happiness puzzles. *American Economic Review, 103*(3), 591–597.

Deci, E. L., & Ryan, R. M. (2000). The "what" and the "why" of goal pursuits: Human needs and the self-determination of behavior. *Psychological Inquiry, 11*, 227–268.

Dolan, P., Layard, R., & Metcalfe, R. (2011). Measuring subjective well-being for public policy. Office for National Statistics.

Duesenberry, J. S. (1949). *Income, saving and the theory of consumer behaviour*. Cambridge: Harvard University Press.

Easterlin, R. A. (1974). Does economic growth improve the human lot? Some empirical evidence. In P. A. David & M. W. Reder (Eds.), *Nations and households in economic growth: Essays in honor of Moses Abramovitz*. New York: Academic Press Inc.

Easterlin, R. A. (2013). Happiness, growth, and public policy. *Economic Inquiry, 51*, 1–15.

Easterlin, R. A., McVey, L. A., Switek, M., Sawangfa, O., & Zweig, J. S. (2010). The happiness-income paradox revisited. *Proceedings of the National Academy of Sciences of the United States of America, 107*, 22463–22468.

FitzRoy, F., Nolan, M. N., Steinhardt, M. F., & Ulph, D. (2014). Testing the tunnel effect: Comparison, age and happiness in UK and German Panels. *IZA Journal of European Labor Studies, 3*(24), 1–30.

Frank, R. H. (2016). *Success and luck: Good fortune and the myth of meritocracy*. Princeton and Oxford: Princeton University Press.

Gagné, M. (2003). The role of autonomy support and autonomy orientation in prosocial behavior engagement. *Motivation and Emotion, 27*(3), 199–223.

Hara, K., Adams, A., Milland, K., Savage, S., Callison-Burch, C., & Bigham, J. (2018). A data-driven analysis of workers' earnings on Amazon Mechanical Turk. In Proceedings of the 2018 ACM conference on human factors in computing systems, arXiv:1712.05796.

Hauser, D., & Schwarz, N. (2016). Attentive Turkers: MTurk participants perform better on online attention checks than do subject pool participants. *Behavior Research Methods, 48*(1), 400–407.

Helliwell, J., Layard, R., & Sachs, J. (2012). *World Happiness Report 2012*. New York: The Earth Institute, Columbia University.

Hirschman, A. O., & Rothschild, M. (1973). The changing tolerance for income inequality in the course of economic development: With a mathematical appendix. *Quarterly Journal of Economics, 87*(4), 544–566.

Ifcher, J., Zarghamee, H., & Graham, C. (2018). Local neighbors as positives, regional neighbors as negatives: Competing channels in the relationship between others' income, health, and happiness. *Journal of Health Economics, 57*, 263–276.

Johansson-Stenman, O., Carlsson, F., & Daruvala, D. (2002). Measuring future grandparents' preferences for equality and relative standing. *Economic Journal, 112*(479), 362–383.

Kahneman, D. (2011). *Thinking fast and slow*. New York: Farrar, Strauss, and Giroux.

Kahneman, D., & Deaton, A. (2010). High income improves evaluation of life but not emotional well-being. *Proceedings of the National Academy of Sciences USA, 107*, 16489–16493.

Kamas, L., & Preston, A. (2015). Can social preferences explain gender differences in economic behavior? *Journal of Economic Behavior and Organization, 116*, 525–539.

Kingdon, G. G., & Knight, J. (2007). Community, comparisons and subjective well-being in a divided society. *Journal of Economic Behavior and Organization, 64*(1), 69–90.

Konow, J., & Earley, J. (2008). The Hedonistic Paradox: Is *homo economicus* happier? *Journal of Public Economics, 92*, 1–33.

Kuhn, P., Kooreman, P., Soetevent, A., & Kapteyn, A. (2011). The effects of lottery prizes on winners and their neighbors: Evidence from the Dutch Postcode Lottery. *American Economic Review, 101*(5), 2226–3224.

Kuziemko, I., Buell, R. W., Reich, T., & Norton, M. I. (2014). "Last-place aversion": Evidence and redistributive implications. *Quarterly Journal of Economics, 129*(1), 105–149.

Kuziemko, I., Norton, M. I., Saez, E., & Stantcheva, S. (2015). How elastic are preferences for redistribution? Evidence from randomized survey experiments. *American Economic Review, 105*(4), 1478–1508.

List, J. A., Shaikh, A. M., & Xu, Y. (2016). Multiple hypothesis testing in experimental economics (No. w21875). National Bureau of Economic Research.

Ludwig, J., Duncan, G. J., Gennetian, L. A., Katz, L. F., Kessler, R. C., Kling, J. R., et al. (2013). Long-term neighborhood effects on low-income families: Evidence from moving to opportunity. *American Economic Review: Papers and Proceedings, 103*(3), 226–231.

Luttmer, E. F. (2005). Neighbors as negatives: Relative earnings and well-being. *Quarterly Journal of Economics, 120*(3), 963–1002.

McBride, M. (2010). Money, happiness: And aspirations: An experimental study. *Journal of Economic Behavior and Organization, 74*, 262–276.

Mogilner, C. (2010). The pursuit of happiness: Time, money, and social connection. *Psychological Science, 21*(9), 1348–1354.

Peterson, R. A., & Sauber, M. (1983). A mood scale for survey research. In P. Murphy et al. (Eds.), *American marketing association educator's proceedings* (pp. 409–414).Chicago: American Marketing Assocations.

Pfeffer, J., & DeVoe, S. E. (2009). Economic evaluation: The effect of money and economics on attitudes about volunteering. *Journal of Economic Psychology, 30*(3), 500–508.

Pigou, A. C. (1920). *The economics of welfare*. London: MacMillan.

Quoidbach, J., Dunn, E. W., Petrides, K. V., & Mikolajczak, M. (2010). Money giveth, money taketh away: The dual effect of money on happiness. *Psychological Science, 21*, 759–763.

Rees, J. C. (1956). *Mill and his early critics*. Leicester: University College.

Senik, C. (2004). When information dominates comparison: learning from Russian subjective panel data. *Journal of Public Economics, 88*(9), 2099–2123.

Senik, C. (2008). Ambition and jealousy: Income interactions in the 'Old' Europe versus the 'New' Europe and the United States. *Economica, 75*(299), 495–513.

Smith, R. H., Diener, E., & Wedell, D. H. (1989). Intrapersonal and social comparison determinants of happiness: A range-frequency analysis. *Journal of Personality and Social Psychology, 56*(3), 317–325.

Snowberg, E., & Yariv, L. (2018). Testing the waters: Behavior across participant pools. In *NBER working paper No. 24781*.

Solnick, S. J., & Hemenway, D. (1998). Is more always better?: A survey on positional concerns. *Journal of Economic Behavior and Organization, 37*(3), 373–383.

Stevenson, B., & Wolfers, J. (2008). Economic growth and subjective well-being. Reassessing the Easterlin paradox. *Brookings Papers on Economic Activity, 1*, 1–102.

Stiglitz, J. E., Sen, A., & Fitoussi, J.-P. (2010). *Report by the commission on the measurement of economic performance and social progress*. Paris: Commission on the Measurement of Economic Performance and Social Progress.

Stone, A., & Mackie, C. (2013). *Subjective well-being: Measuring happiness, suffering, and other dimensions of experience*. Washington, DC: The National Academies Press.

Vohs, K. D., Mead, N. L., & Goode, M. R. (2006). The psychological consequences of money. *Science, 314*(5802), 1154–1156.

Zaleskiewicz, T., Gasiorowska, A., Kesebir, P., Luszczynska, A., & Pyszczynski, T. (2013). Money and the fear of death: The symbolic power of money as an existential anxiety buffer. *Journal of Economic Psychology, 36*, 55–67.

Zeckhauser, R. J. (1991). The strategy of choice. In R. J. Zeckhauser (Ed.), *Strategy and choice* (pp. 1–21). Cambridge, MA: MIT Press.

Zhou, X., Vohs, K. D., & Baumeister, R. F. (2009). The symbolic power of money: Reminders of money alter social distress and physical pain. *Psychological Science, 20*(6), 700–706.

**Publisher's Note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.